

Research Statement

Regina (Jeanne) Ruane

My research lies at the intersection of data science, network analysis, and digital experimentation, with a central focus on the dynamics of participation, learning, and influence within complex, networked systems. My trajectory—from online education and peer mentoring to healthcare information technology and FinTech—reflects a consistent effort to understand how social, spatial, and temporal structures shape individual and collective behavior in digitally mediated environments.

As Managing Director of the Computational Social Science Lab at the University of Pennsylvania, I lead a research portfolio focused on the design and analysis of digital experiments involving business analytics, network inference, and data-driven modeling of human behavior. This work spans sectors including financial technology, crowdfunding, and emerging innovation ecosystems, and is driven by strong collaborations with industry partners such as Wells Fargo, Microsoft Research, Nielsen, Amazon, Blackrock, Disney, as well as a range of startups.

A defining theme across my research is methodological innovation in analyzing network data arising from complex observational studies. My early work examined online peer mentoring systems in educational settings, using a mixed-methods approach that combined social network analysis, interpretive content analysis, and critical discourse analysis. These studies provided insight into the formation of online learning communities and the role of social positioning, peer influence, and discourse in promoting student engagement and retention. This research received recognition in journals such as *Online Learning* (Best Paper Award) and *Journal of Online Learning and Teaching*.

Subsequent work expanded into healthcare information technology, where I applied network and content analysis to understand the structure and use of patient portal systems. A systematic review of ten years of HIT literature, conducted with collaborators at the University of Illinois, mapped the thematic and methodological evolution of the field and identified key research gaps—particularly in underexplored populations and technologies such as radiology information systems and virtual reality.

Currently, my research focuses on high-resolution modeling of networked behavior in the domains of FinTech and AI systems. I work with applied researchers and engineers in both academic and corporate environments to design and evaluate digital interventions using large-scale network data. I am particularly interested in sampling and inference under complex dependence, network causal inference, and learning dynamics in platform-based ecosystems. These research questions are motivated by practical

problems in governance, fairness, and innovation, and contribute to both methodological development and applied policy insight.

I have held prior appointments as Executive Director of the Data Science Institute and Assistant Professor at Temple University, collaborating with researchers at institutions including Columbia, MIT, Stanford, the University of Chicago, and Yale, as well as with major technology firms such as Amazon, Google, and Disney. My academic background includes a PhD from Drexel University, an MA from Columbia University, and a BA from Villanova University.

Looking forward, I am committed to expanding the frontier of data science through interdisciplinary collaboration, cutting-edge analytics, Artificial Intelligence, and a focus on real-world impact. I seek to continue developing research that bridges data science, social systems, and technology—with the aim of understanding, predicting, and improving collective outcomes in digitally networked environments.